THE HONG KONG POLYTECHNIC UNIVERSITY
AAE4203

Teaching Plan
AAE4203 Guidance and Navigation
Academic Year 2026-2027 Semester 1
Department of Aeronautical and Aviation Engineering

1 Course Information

Course Code: AAE4203
Course Title: Guidance and Navigation
Credit Units: 3
Lecture Time: Friday 3:30 pm - 6:30 pm (venue: CD302)
Instructor: Prof. Weisong WEN (welson.wen@polyu.edu.hk)
Teaching Assistants: Mr. Jiahao Hu, Mr. Yingming Chen, Mr. Xupeng Zhai

2 Subject Intended Learning Outcomes (SILOs)

Upon completion of this subject, students will be able to:

a. Understand and explain the working principles of navigation and guidance systems for unmanned autonomous systems (UAS); and

b. Competently apply the fundamental mathematical concepts of UAS navigation; and

c. Critically evaluate the characteristics, purposes, and design procedures of UAS navigation and guidance systems; and

Identify the technological and design trends of future UAS navigation.

3 Assessment Scheme

Assessment Component	**Weighting**
Assignment 1	7.5%
Assignment 2	7.5%
Mid-term Test	15%
Group Project	20%
Final Examination	50%
Total	**100%**

4 Student Study Effort Expected

Class contact:		
	Lecture	30 Hrs.
	Laboratory/Tutorial	6 Hrs.
Other student study effort:		
	Week 3 Reading Week	3 Hrs.
	Continuous Assessment	35 Hrs.
	Self-study	36 Hrs.
	Total student study effort	**110 Hrs.**

5 Teaching Schedule

Week	Date	Topics
1	Sept 4	Mathematical Foundations for Guidance and Navigation - Part 1
2	Sept 11	Mathematical Foundations for Guidance and Navigation - Part 2
3	Sept 18	Reading Week - No Class
4	Sept 25	GNSS Fundamentals - Part 1
5	Oct 2	GNSS Single Point Positioning (SPP) - Part 1
6	Oct 9	GNSS Single Point Positioning (SPP) - Part 2
7	Oct 16	GNSS Velocity Estimation
8	Oct 23	Mid-term Test
9	Oct 30	Laboratory: GNSS Positioning and Velocity Estimation
10	Nov 6	GNSS RTK and High-Precision GNSS Positioning
11	Nov 13	Group Project Presentations
12	Nov 20	Group Project Presentations
13	Nov 27	Tutorial and Revision

Note: Week 3 is designated as a reading week without an in-class meeting. Week 13 is reserved for tutorial and revision to consolidate the course material before the final examination.

6 Required Textbooks

1. Farrell, J.A. (2008). Aided Navigation: GPS with High Rate Sensors. McGraw-Hill.
2. Groves, P.D. (2013). Principles of GNSS, Inertial, and Multisensor Integrated Navigation Systems. Artech House.

7 Course Policies

- Attendance: Regular attendance is expected for all classes.
- Late Submission: A 10% penalty per day applies to late assignments.
- Academic Integrity: All submitted work must be original and comply with PolyU requirements.
- Students must attend the final examination in order to attain a passing grade in the overall result

This teaching plan is subject to change. Version date: August 17, 2026